DIFFERENCES OF DIVISORS ON A CURVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 725-732 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 734-750, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 18 of queue_pic.json (pb=826, stmt 725-732, proof 734-750).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0B8V: sources/stacks-pic/pinned-original-source.tex, source lines 750–750. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9D: sources/stacks-pic/pinned-original-source.tex, source lines 625–634. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9E: sources/stacks-pic/pinned-original-source.tex, source lines 678–683. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02KB: sources/stacks-pic/pinned-original-source.tex, source lines 750–750. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $X \to S$ be a smooth morphism of schemes of relative dimension 1. Let $D_1, D_2 \subset X$ be closed subschemes finite locally free of degrees d_1, d_2 over S. If $D_1 \subset D_2$ (as closed subschemes) then there is a closed subscheme $D \subset X$ finite locally free of degree $d_2 - d_1$ over S such that $D_2 = D_1 + D$.*

Proof. This proof is almost exactly the same as the proof of Lemma 0B9E. By Lemma 0B9D we see that D_1 and D_2 are relative effective Cartier divisors on X/S. By Divisors, Lemma 0B8V there is a relative effective Cartier divisor $D \subset X$ such that $D_2 = D_1 + D$. Hence $D \to S$ is locally quasi-finite, flat, and locally of finite presentation by Lemma 0B9D. Since D is a closed subscheme of D_2, we see that $D \to S$ is finite. It follows that $D \to S$ is finite locally free (Morphisms, Lemma 02KB). Thus it suffice to show that the degree of $D \to S$ is $d_2 - d_1$. This follows from Lemma 0B9E. □